



Module-3: Work & Heat

Thermodynamics (NMEC103) : (Winter 2024-25)

Dr. S. Sahoo, Mechanical Engineering, IIT Dhanbad

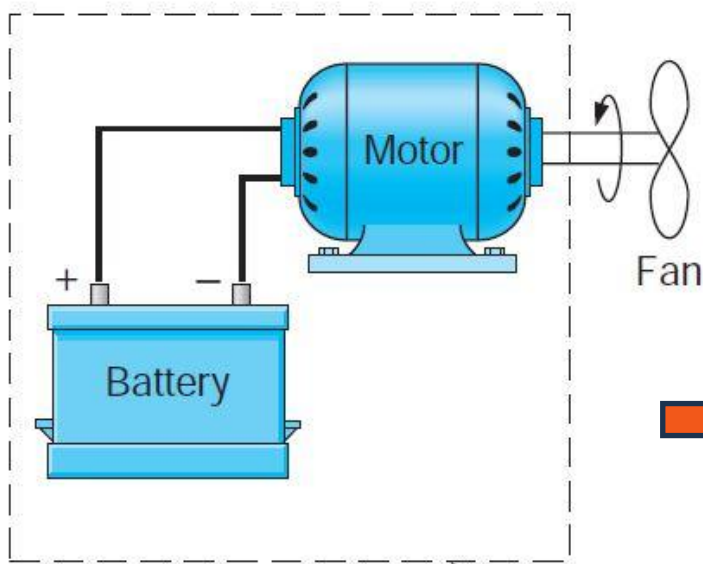


Fig. a

System boundary

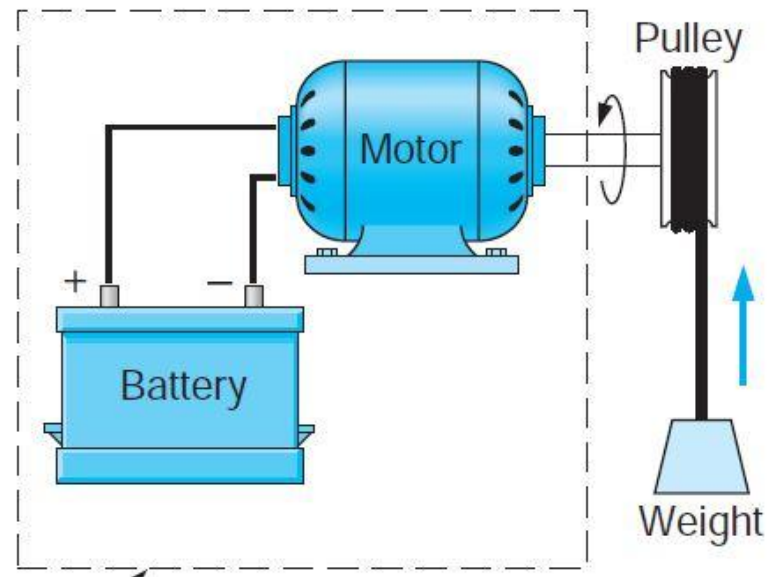


Fig. b

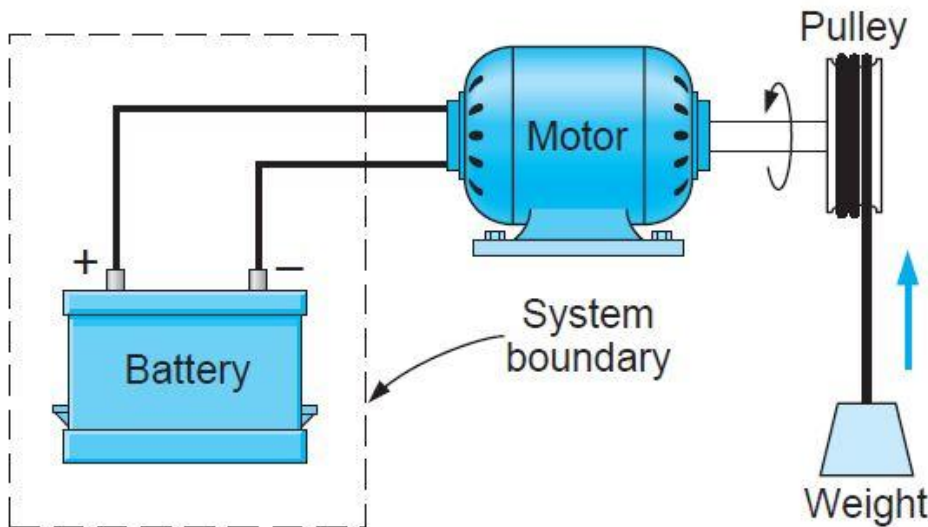
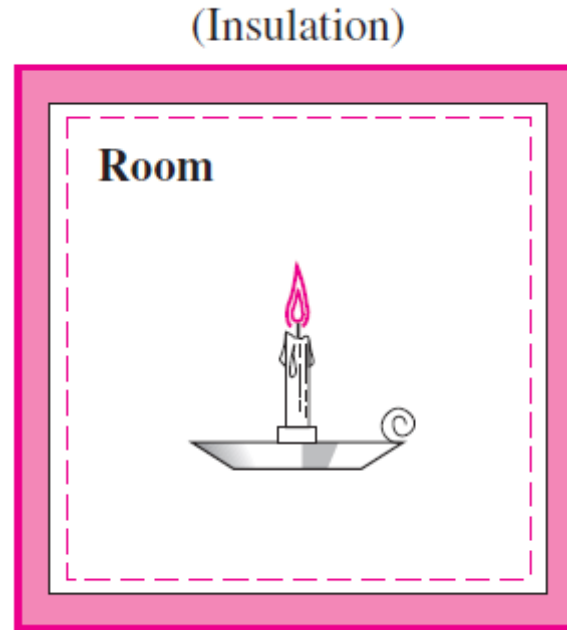


Fig. c

The only limiting factor when the sole external effect is the raising of a weight is the inefficiency of the motor. However as we design a more efficient motor, with lower bearing and electrical losses, we recognise that we can approach a certain limit that meets the requirement of having the only external effect be the raising of a weight. Therefore we can conclude that when there is flow of electricity across the boundary of system (Fig. c), it is work

Example: A candle is burning in a well-insulated room. Taking the room (the air plus the candle) as the system, determine (a) if there is any heat transfer during this burning process and (b) if there is any change in the internal energy of the system.



We have an adiabatic system and no heat will pass through the boundaries. Therefore, $Q = 0$ for this process. The internal energy involves energies that exist in various forms (sensible, latent, chemical, nuclear). During the process just described, part of the chemical energy is converted to sensible energy. Since there is no increase or decrease in the total internal energy of the system, $\Delta U = 0$ for this process.

➤ In **mechanics**, work is defined as the product of force and component of the displacement parallel to the force, given by the scalar product. $W = \int \mathbf{F} \cdot d\mathbf{s}$ Here “s” corresponds to the displacement of the point of application of the force. On the other hand, thermodynamic definition of work represents an energy interaction between the system and the surroundings (external work). Thus a gas confined in a cylinder and at uniform pressure, while expanding and imparting motion to a piston does external work on the surroundings. The work done, however, by one part of the system on another part is called internal work. The interaction of molecules, atoms or electrons on one another constitute internal work. Internal work can not be discussed in macroscopic thermodynamics.

➤ When a system transfers energy in the form of work, it does positive work. At the same time, surroundings will do an equal amount of negative work.

➤ A very important aspect of work interaction is that the process carried out by the system, must be resisted by the surroundings. If the surroundings do not offer any resistance, then the work done by the system will become zero.

➤ Free fall of a mass

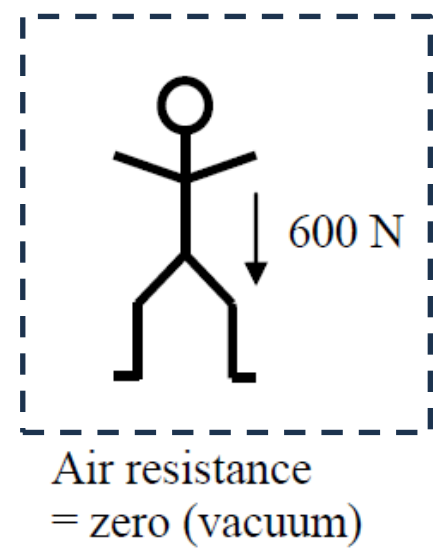
According to the Mechanics definition, the work done during this process $W = mgh$. However, according to the Thermodynamic definition, work done by the mass $W_{mass} = 0$. In this case of free fall, although the force of body weight ($= mg$) moves the centre of gravity of the mass by a distance of h , actually no energy crosses the system boundary. An alternative way of understanding this example is to consider the fact that the surrounding vacuum does not offer any resistance to the motion of the mass; work exchange between the system and surroundings is possible only when the process is resisted by the surroundings.

Example 1: Free fall of a mass

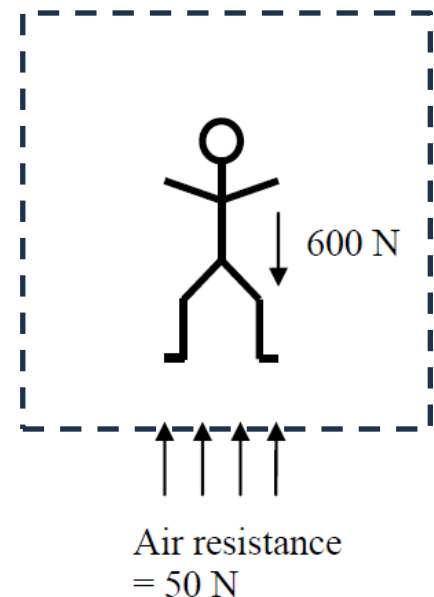
Let us now consider the free fall of a particle of mass “m” in vacuum, under the action of gravity. If the mass falls by a height “h”, according to the Mechanics definition, the work done during this process $W = m \cdot g \cdot h$. However, according to the Thermodynamic definition, work done by the mass $W_{mass} = 0$. In this case of free fall, although the force of body weight ($= mg$) moves the center of gravity of the mass by a distance of h, actually no energy crosses the system boundary. As per the Thermodynamic definition, work is an energy interaction between the system (or CV) and the surroundings. Here, no energy interaction took place- only there was some internal energy conversion for the mass. The potential energy of the mass at higher elevation (initial position of the mass) was converted into its own kinetic energy as the mass accelerated during the fall. It is similar to a situation when a person takes out a 100 rupee note from one pocket, gets it converted to 10 notes of ten rupees and puts them in another pocket- in other words, there is money exchange in this transaction. In the case of the free-falling mass, the total energy of the mass, i.e. potential energy + kinetic energy, remains the same. Since there is no energy interaction, work done has to be zero. *An alternative way of understanding this example is to consider the fact that the surrounding vacuum does not offer any resistance to the motion of the mass; work exchange between the system and surroundings is possible only when the process is resisted by the surroundings.*

Resisted/ unresisted fall under the action of gravity

In order to illustrate the idea of **fully-resisted**, **partially resisted** and **unresisted** processes let us consider another example. In the event of an aircraft emergency, a passenger of **60 kg weight** (**~ 600 N force**) decides to jump out of the aircraft and we are interested to find out the work done when the person descends by a height of **100 m**. Let us consider three cases, namely: (a) **when there is no air resistance** (b) **when air resistance does not fully meet the force due to the weight of the person** (c) **when the person ties himself onto a parachute and descends very slowly**. In cases (a) and (b), since weight is not equal to air resistance, the person will accelerate. In case (c) when air resistance is equal to the weight, the **person will descend slowly without any acceleration**. As regards the amount of work done by the person, in case (a) work = zero, since no resistance is offered by the vacuum. In case (b), work done = air resistance x distance of fall = $50 \text{ N} \times 100 \text{ m} = 5000 \text{ J} = 5 \text{ kJ}$.

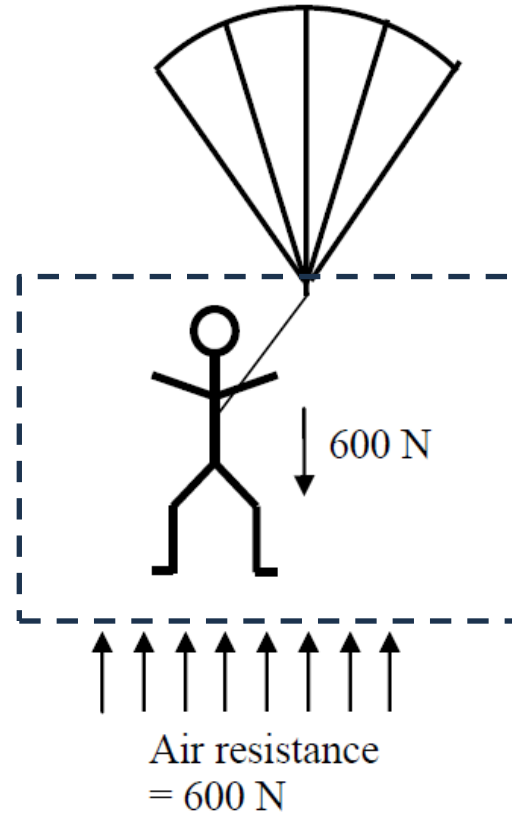


a) Unrestricted



b) Partially Resisted

In case (c), work done = air resistance x distance of fall = weight x distance of fall = $600 \text{ N} \times 100 \text{ m} = 60000 \text{ J} = 60 \text{ kJ}$.



c) Fully restricted

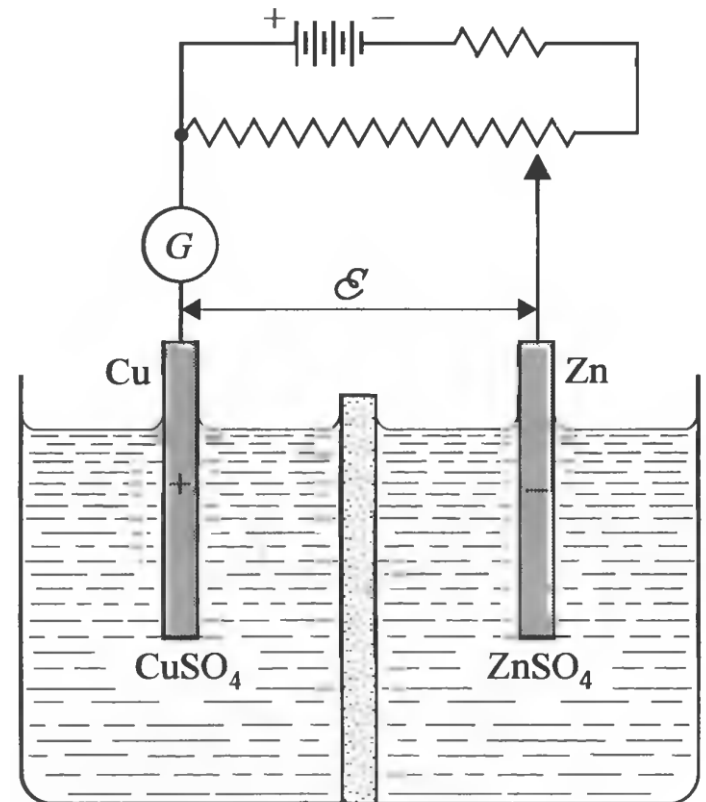
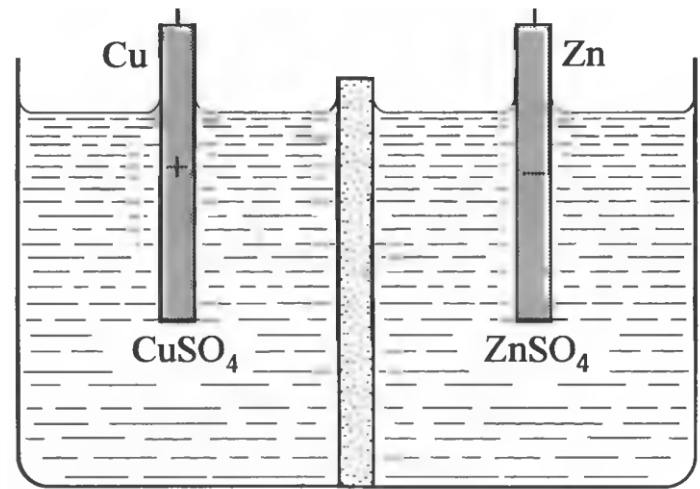
Some Important Points Regarding Work

- Work is an **energy in transit**
- Work **cannot be stored** by system or control volume
- Work is a form of energy which can **only be identified at the system boundary/ control surface**.
- Work **loses its identity once it crosses the system boundary**
- If the surrounding **won't give any resistance** the work done by the system on the surrounding will be zero
- **Thermodynamic work** is **somehow different from the definition of work given in mechanics**.
- Unlike the thermodynamic properties **work is a path function**, hence depends on **initial, final states as well as the path** followed by the system to go from initial to final state.
- **Area under the path on thermodynamic platform gives work for a reversible process**.
- For **irreversible process** the path is not defined, so the associated work can not be calculated from area concept.

- The differential of work is inexact in nature hence is represented as δW instead of dw . δW is not the differential of W .
- For simple compressible pure system, the only mode of work is the moving boundary work. If it is simple compressible but multi component system additionally the chemical work associated with diffusion of species will come into picture
- Work is not a property of the system.
- Work transfer is due to the organized motion in the surrounding.
- Work is a high-grade energy hence more costly.

➤ Electrochemical cell:

If an electrochemical cell is disconnected from the circuit, changes that take place in the cell (such as diffusion) are not accompanied by the performance of work. If, however the cell is connected to an external circuit through which charges flow, the current may be imagined to produce the rotation of the armature of a motor, thereby lifting weight or winding a spring. **Therefore, for an electrochemical cell to do work it must be connected to an external circuit.**



➤ Magnetization:

A change of magnetization within the magnet is not accompanied by the performance of work. If, however, the magnet undergoes a change of magnetization while it is surrounded by an electric conductor, then eddy currents are set up in the conductor, constituting an external transfer of charge. *Hence for a magnetic system to do work, it must interact with external electric conductor or with other magnets.*

Mechanical Work Modes of Energy Transport

In Mechanics we recognise that work is done whenever a force moves through a distance. When ever this force is a mechanical force F we call this work mode *Mechanical Work* and is defined as

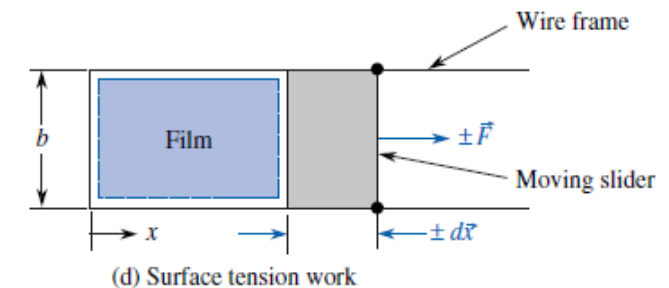
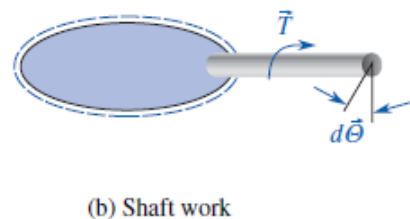
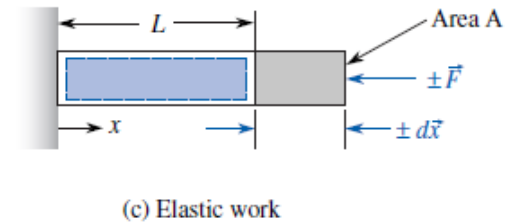
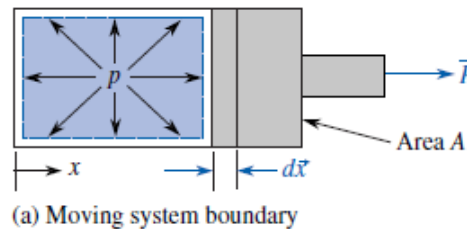
$$(\delta W)_{\text{Mechanical}} = \mathbf{F} \cdot d\mathbf{x}$$

$$(W)_{1-2, \text{Mechanical}} = \int_{x_1}^{x_2} \mathbf{F} \cdot d\mathbf{x}$$

Where the force F is assumed to be exerted by rather on the system. Also our sign convention requires that work done by the system be positive, while work done on the system is negative

In thermodynamics there are four classical types of mechanical work

- 1) Moving system boundary work
- 2) Rotating shaft work
- 3) Elastic work
- 4) Surface tension work

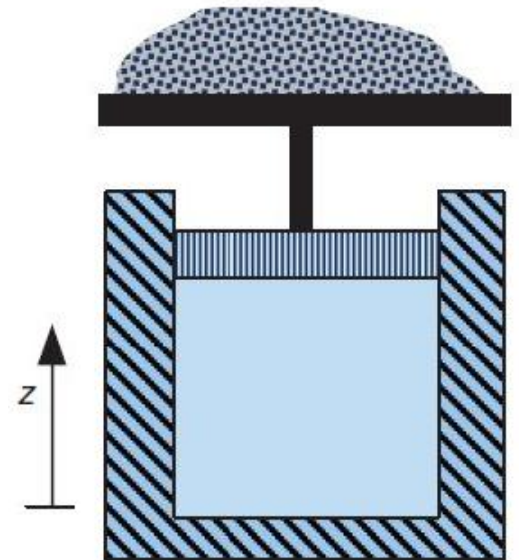


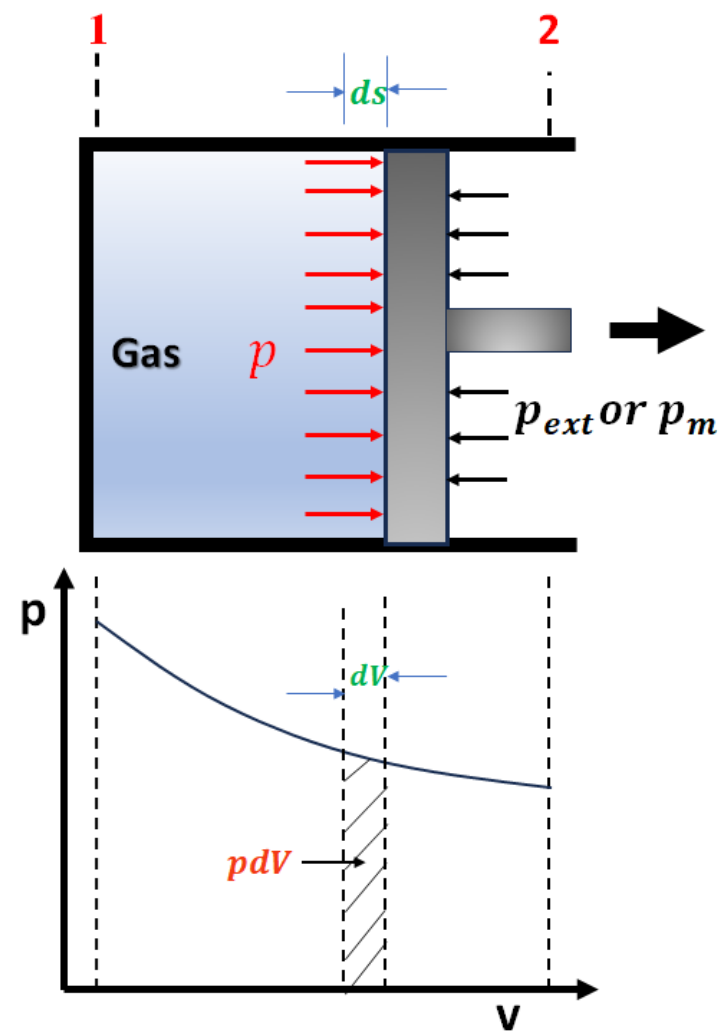
Work of a Frictionless Quasiequilibrium Process of a Simple Compressible Closed System

- ❖ A simple compressible substance involves no effects of electricity, magnetism, anisotropic stress. Or surface tension (An incompressible fluid is a special case in which the density is independent of pressure (in a loose sense density is constant)) A simple compressible substance and involve no effects of motion or gravity is called simple compressible closed system
- ❖ For example, if we are told that air at 300 kPa, 30°C, is stored in a 1-m diameter spherical tank, we generally do not ask where in the tank the pressure is 300 kPa. The reason we do not ask is that the difference in pressure between air at the highest point and the lowest point is only about 0.3 kPa, and for most purposes the slight variation of pressure due to gravity can be neglected



- ❖ The effects of motion such as the acceleration of part of the system by means of a moving pistons are often negligible
- ❖ There are processes in which the system is essentially in equilibrium at any instant. In the case of the gas in the cylinder, if we make sure that the external equivalent pressure is equal (or essentially equal) to the pressure of the gas throughout the expansion process, then at any instant during the process the gas could be in equilibrium. A process in which the system passes through equilibrium states during a change of state is called a *quasistatic process*. The work of a quasistatic process is called *quasistatic work*.





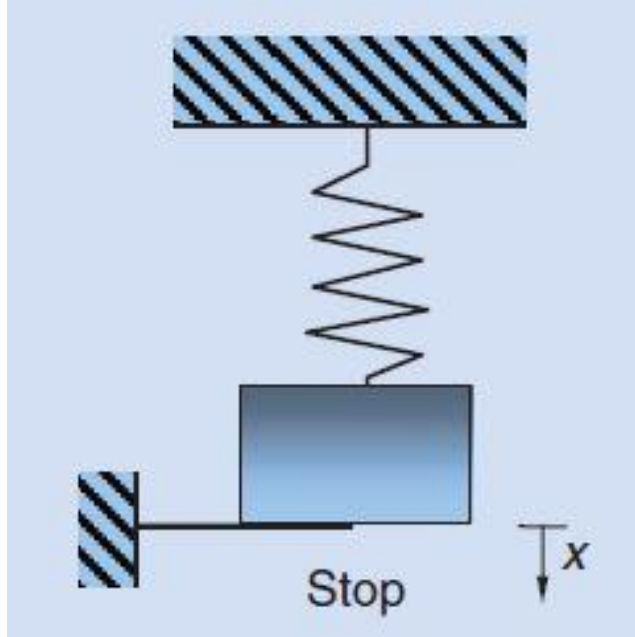
To develop an expression for the work of **Quasiequilibrium process of a simple compressible closed system**. Consider a gas trapped in a cylinder and expanding against a piston as shown in the figure. The gas expands from state 1 to 2. **The piston moves slowly so that the effect of motion on the system are negligible.** The pressure is uniform throughout the system at any stage of expansion, but the value of this expansion pressure varies as the expansion proceeds. At any stage of expansion, the force on the piston is the product of pressure of the gas and the area of the piston. Since this force act in the direction of motion of the piston, the work done by the gas on the piston while the piston moves a distance ds is

$$\delta W = F ds = p A ds = p dV$$

$$W_{1-2} = \int_1^2 p dV \quad w_{1-2} = \int_1^2 p dv$$

$$W_{1-2} = \int_1^2 -p_{ext} dV \leq \int_1^2 p dV$$

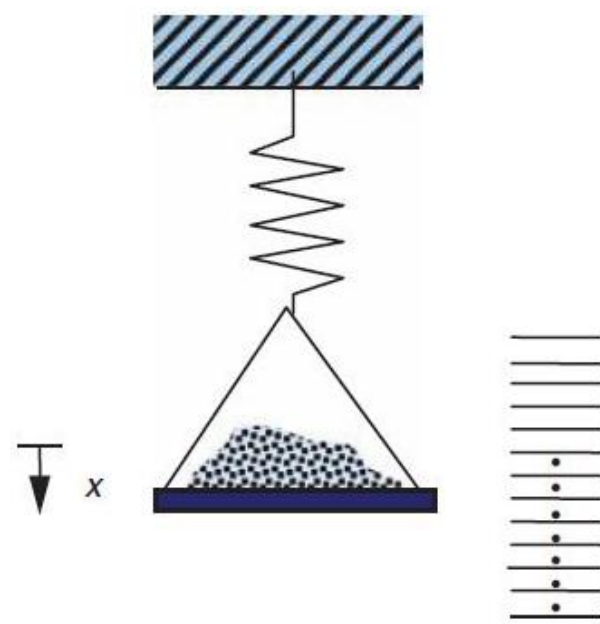
Note that in general the work may be equal to or less than the integral



A 5 kg block is attached to a linear spring and is resting on a support. The spring is under zero tension. When the support is removed, the block drops and after several oscillations it assumes a new position. If the spring constant is $k = 500 \text{ N/m}$, calculate the work interaction of the spring, neglecting the resistance of air.

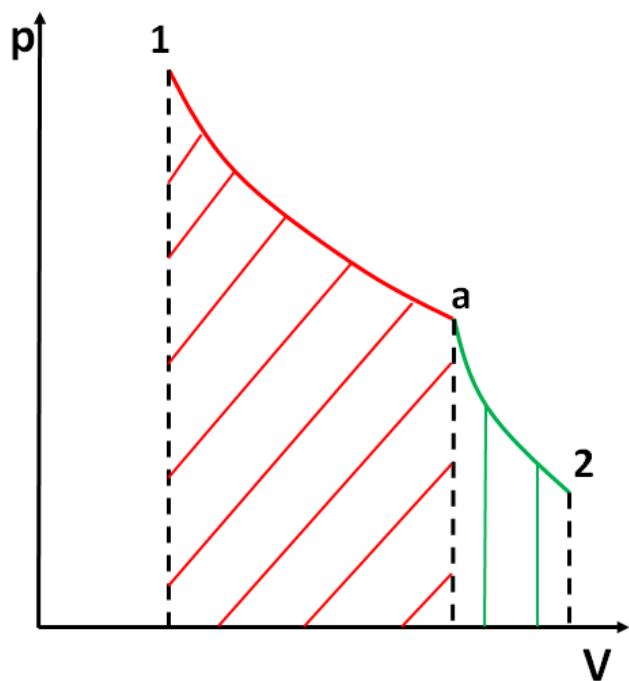
$$W = -mgx$$

$$kx = mg$$

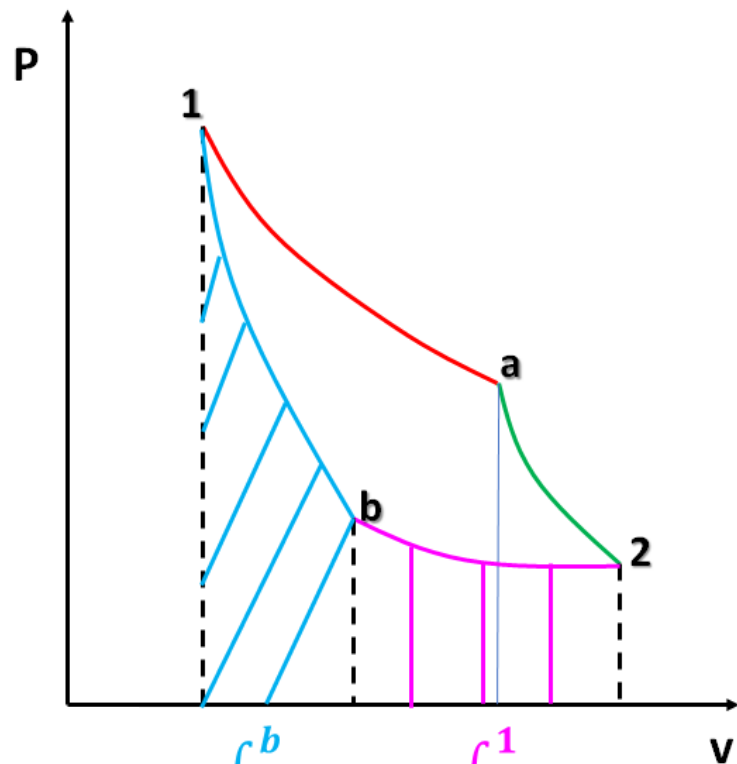


The process of stretching the spring can be performed quasistatically by starting with zero load and increasing the tension on the spring by continuously adding fine grains of sand to the load as shown in Figure. At any instant the spring is in mutual equilibrium with the load and therefore the process is quasistatic, and the work can be calculated from

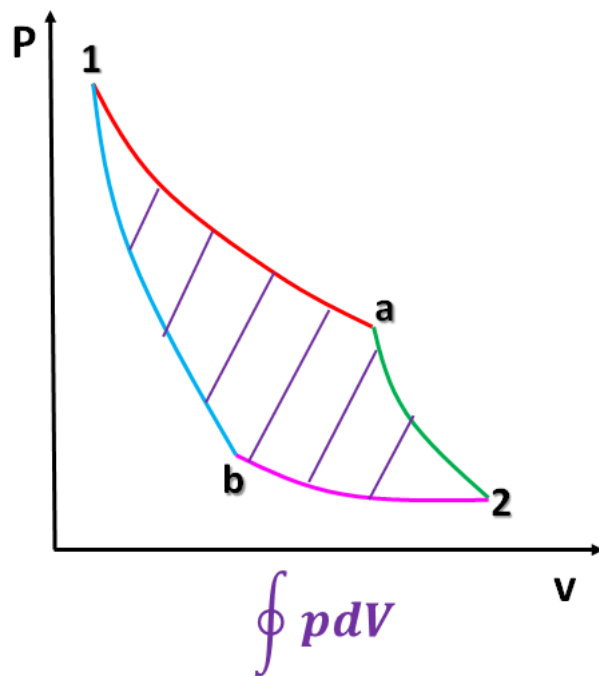
$$W = \int_1^2 F dx = - \int_1^2 kx dx = -\frac{k}{2}(x_2^2 - x_1^2) = -\frac{mgx}{2}$$



$$\int_1^a p dV + \int_a^2 p dV$$



$$\int_2^b p dv + \int_b^1 p dv$$

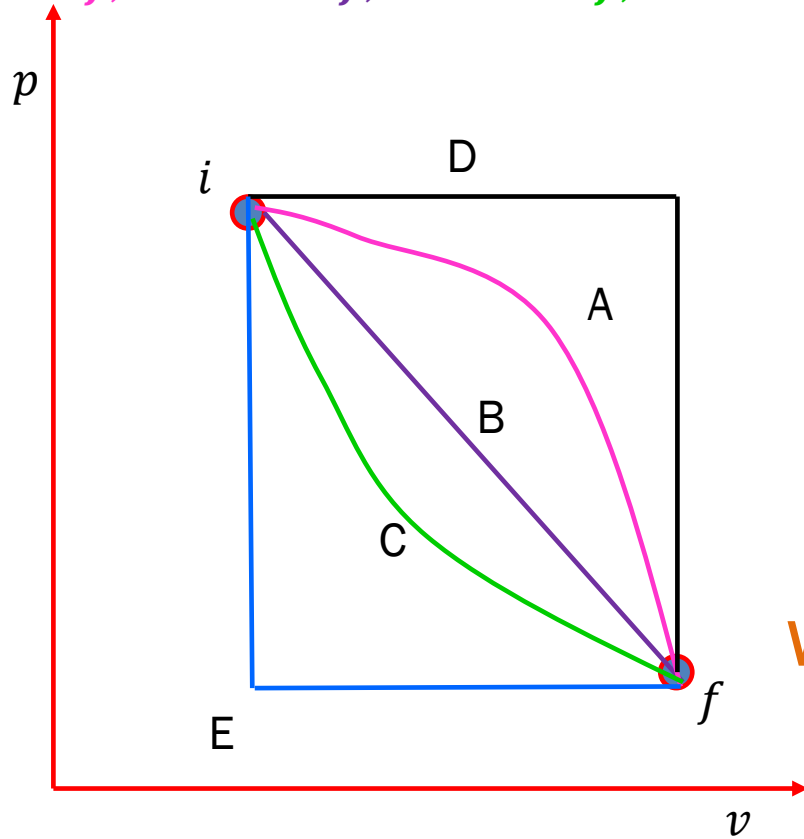


$$\oint p dv$$

Check

$$\oint p dv = \oint v dp$$

$$W_{i-f,A} \neq W_{i-f,B} \neq W_{i-f,C} \neq W_{i-f,D} \neq W_{i-f,E}$$



Work Is a path function

$$\delta w_{rev,nonflow} = p dv + 0 dP$$

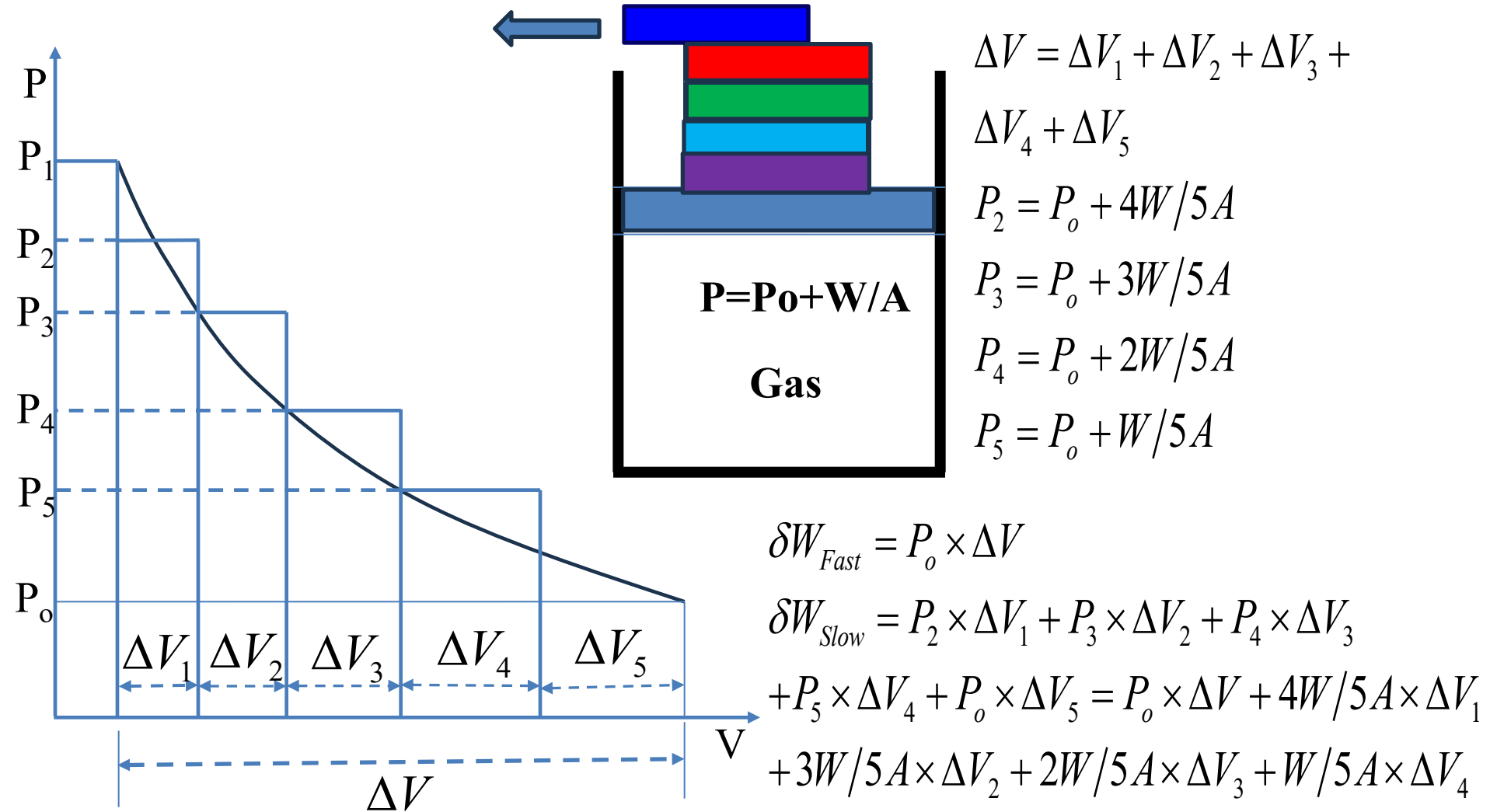
$$\delta w_{rev,nonflow} = M dx + N dy$$

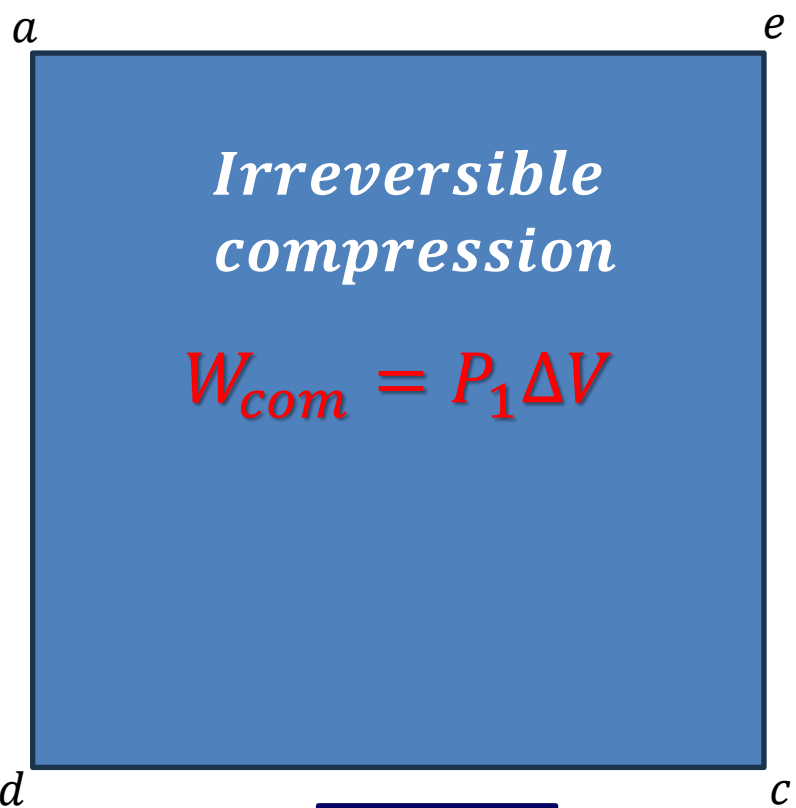
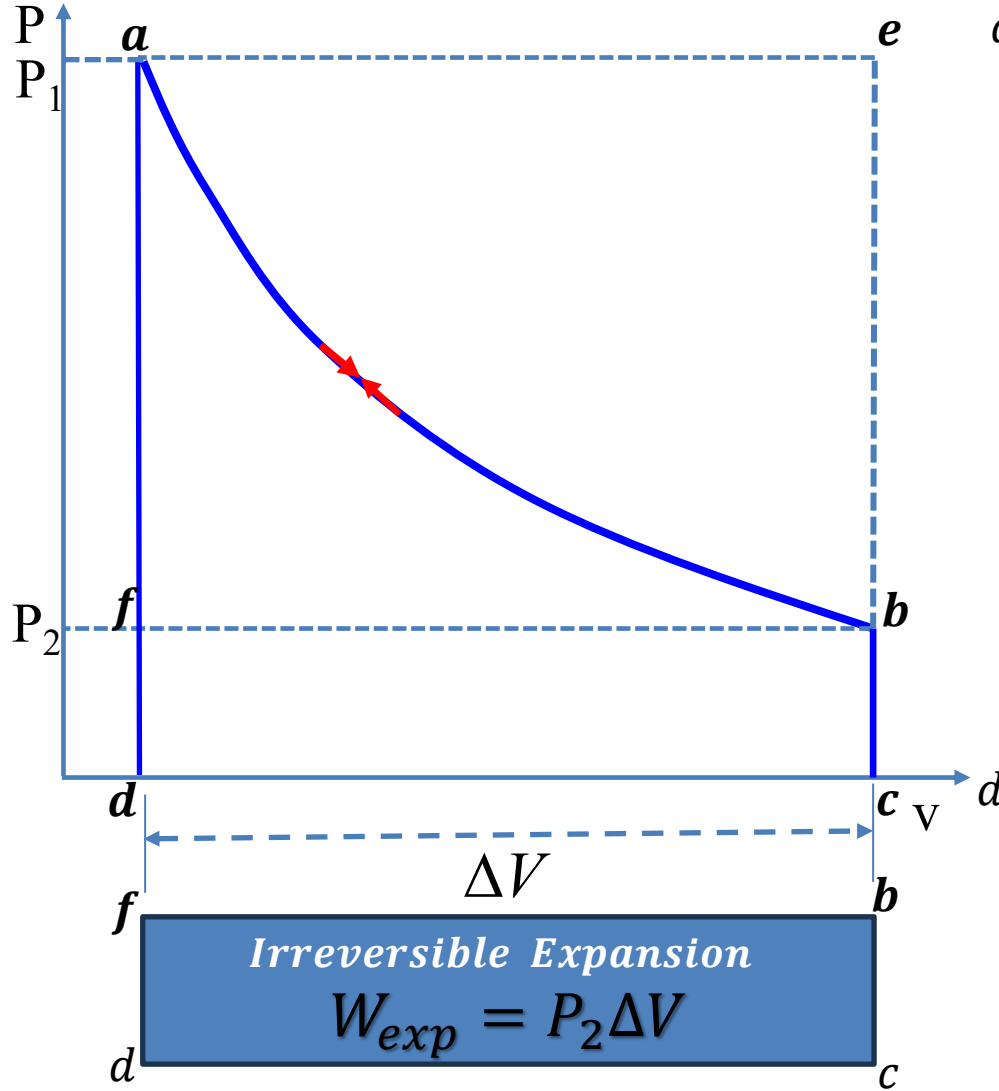
$$M = p; x = v; N = 0; y = p$$

$$\frac{\partial M}{\partial y} = \frac{\partial p}{\partial p} = 1; \frac{\partial N}{\partial x} = \frac{\partial 0}{\partial v} = 0$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \Rightarrow \delta w \text{ is an inexact differential (path function)}$$

How does fast process give rise to loss of work potential?



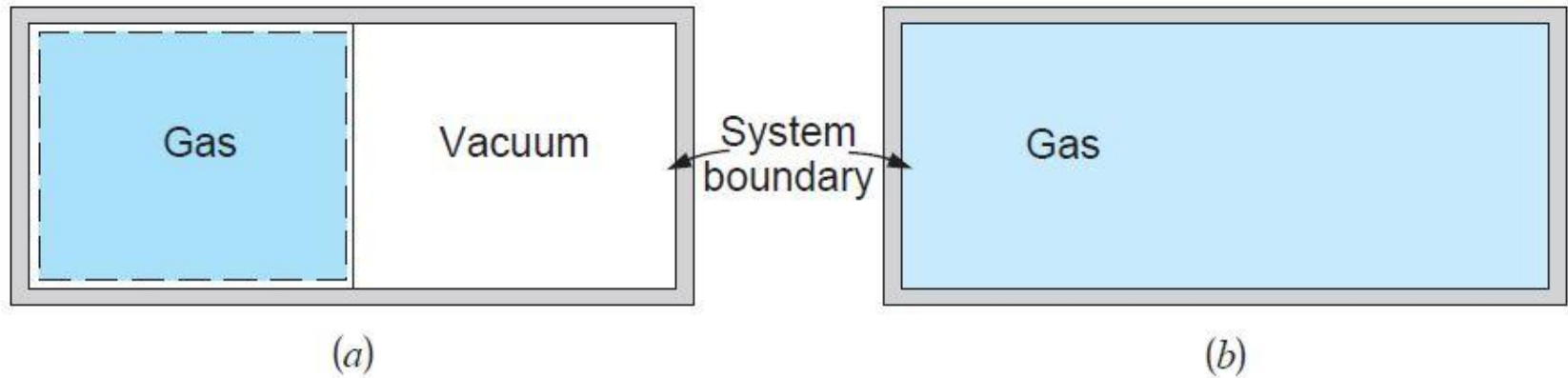


Work of compression reversible: $b - a - d - c - b$

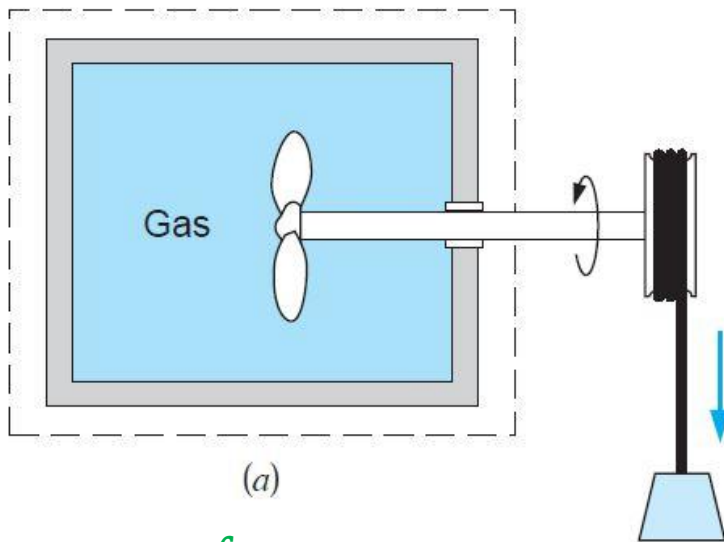
Work of expansion reversible: $a - b - v - d - a$

Free Expansion

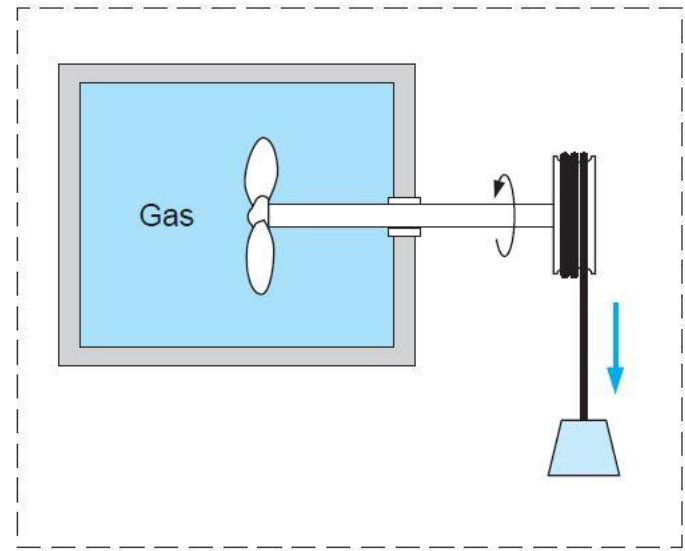
It is important to note that $\int p dV$ is not always equal to work done



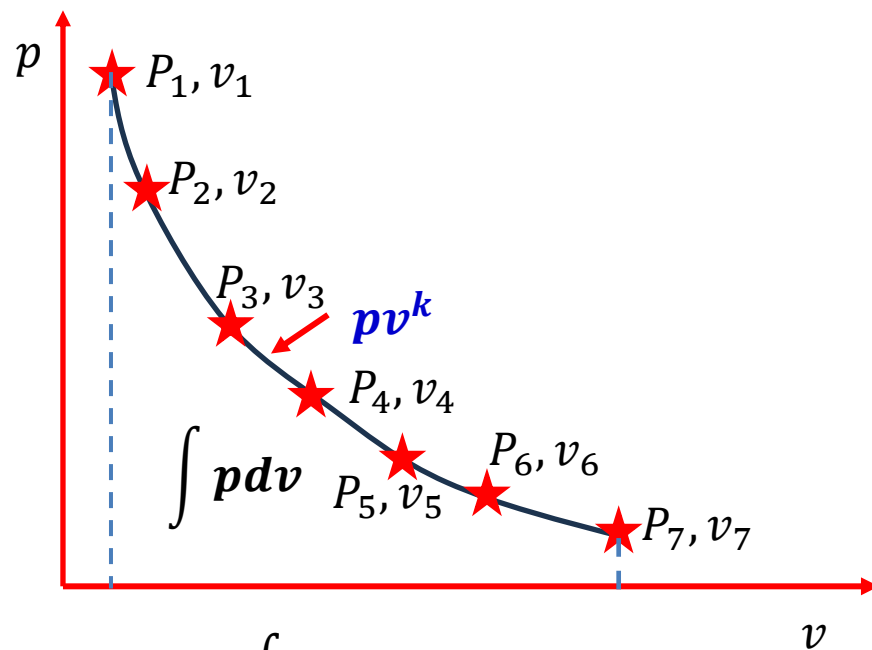
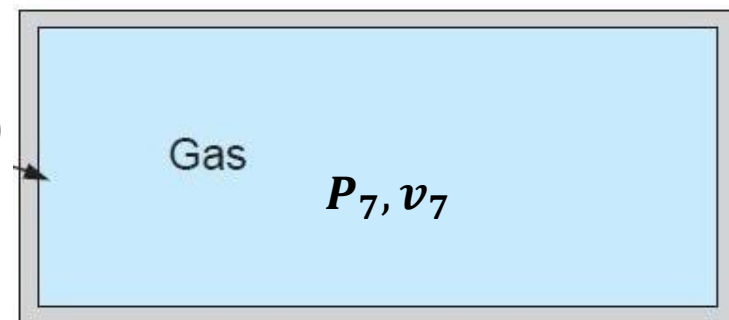
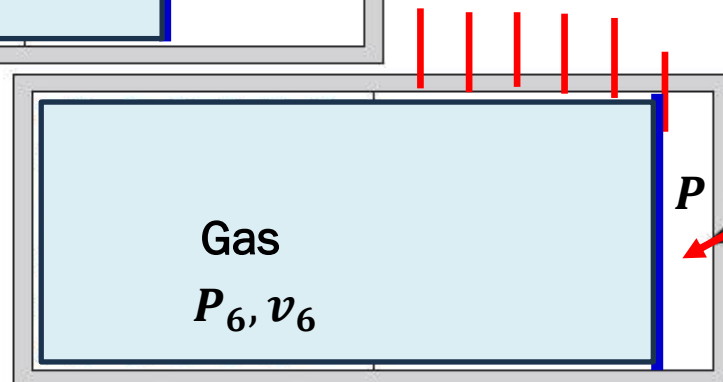
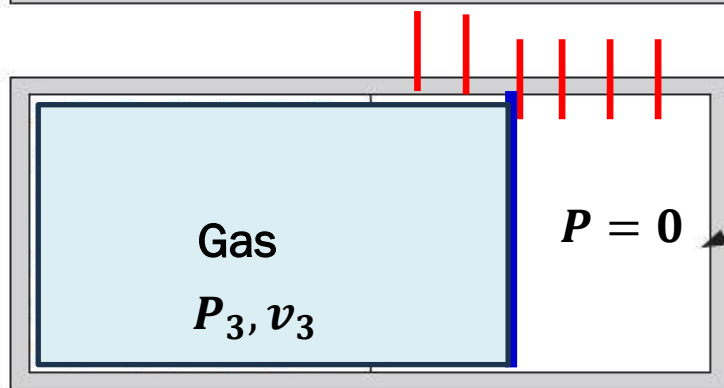
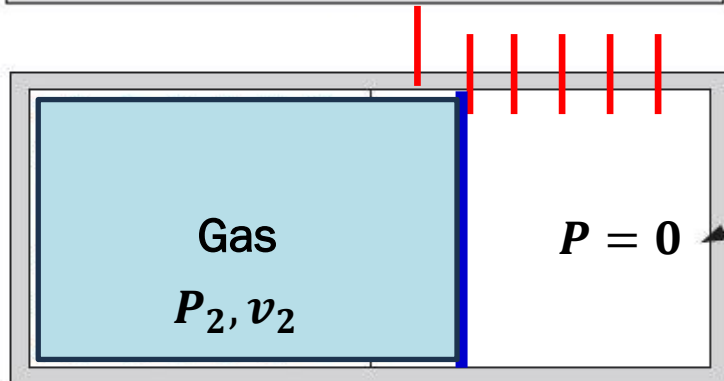
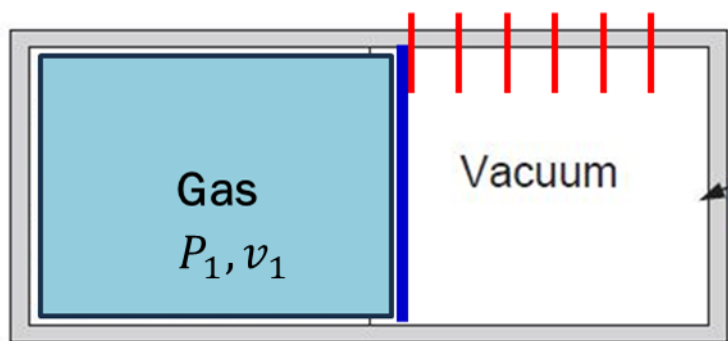
$$W = 0 \quad \int p dV \neq 0 \quad W \neq \int p dV$$



$$W \neq \int p dV \quad W < 0$$



$$W = 0$$



$$\int p dv \neq w_{1-2} = 0$$

Moving Boundary Work of a Quasiequilibrium and frictionless (Reversible) process of a simple compressible closed system / substance in steady flow

$$W_{nonflow,rev} = \int FAdx = \int pdV$$

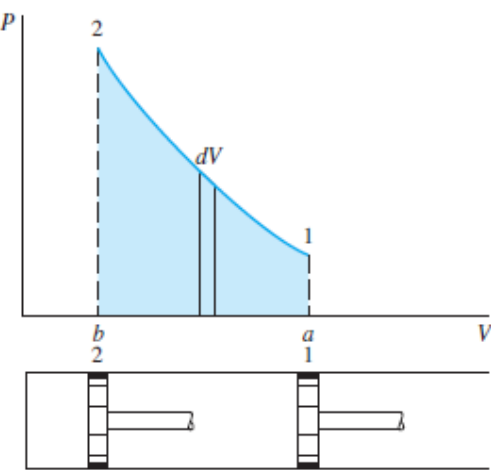
The general form for work in reversible process for closed system

$$\delta W = (\text{intensive property}) d(\text{extensive property})$$

There are different mechanical form of work (shaft work, elastic work, acceleration work, surface tension work, spring work), non-mechanical form of work (electrical work, magnetic work, chemical work)

We can write a general expression for work in terms of some generalised force and displacement term

$$W_{1-2} = \int_1^2 F_{gen} x_{gen}$$



Note: We should also realize that there may be boundary movement work in a nonequilibrium process. Then the total force exerted on the piston by the gas inside the cylinder, PA , does not equal the external force, F_{ext} , and the work is not given by Eq. $W = \int p dv$. The work can, however, be evaluated in terms of F_{ext} or, dividing by area, an equivalent external pressure, P_{ext} . The work done at the moving boundary in this case is:

$$\delta w = P_{\text{ext}} dV$$

Note: Evaluation of above equation in any particular instance requires a knowledge of how the external force or pressure changes during the process.

Note: When a system does an external work the changes that takes place can be described by means of macroscopic quantities referring to the system as a whole

$$W_{nonflow,rev} = \int p dV$$

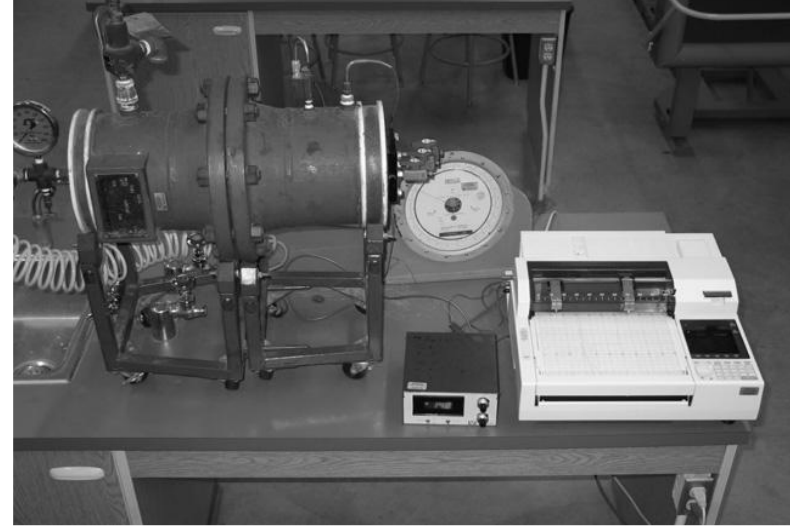
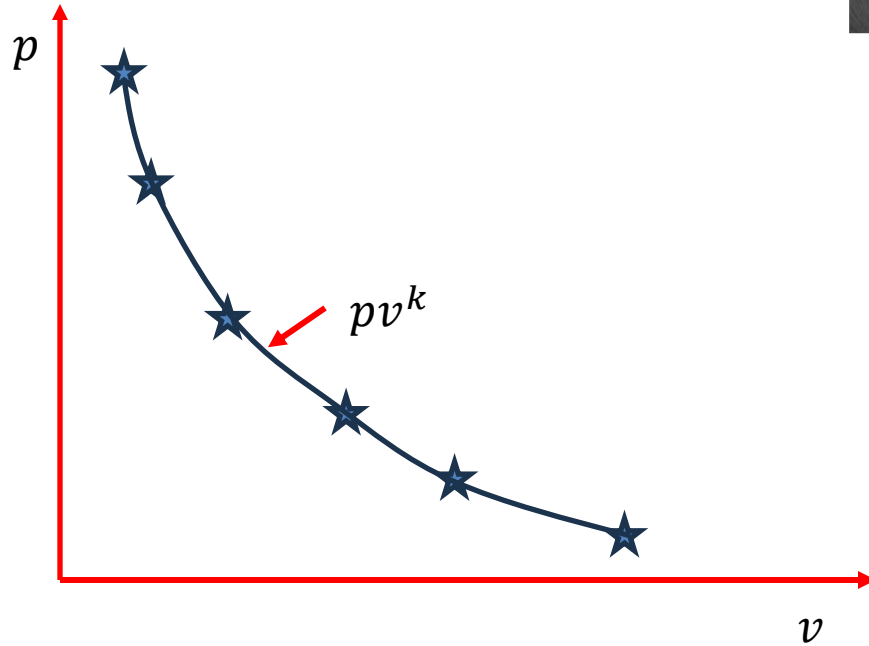
In evaluating the integral of the above equation, we should always keep in mind that we wish to determine the area under the curve as shown in the figure. In connection with this point, we identify the following two classes of problem.

1. The relationship between P and V is given in terms of the experimental data or in graphical form (as, for example, the trace on an oscilloscope). Therefore, we may evaluate the integral by graphical or numerical method.
2. The relationship between P and V makes it possible to fit an analytical relationship between them. We may then integrate directly.

Polytropic Process

$$pV^k = \text{constant} \quad -\infty \leq k \leq \infty$$

$$p_i V_i^k = p_f V_f^k = \text{constant}$$



Use actual data from the experiment shown here to find the **polytropic exponent** for expanding air

Polytropic Process

$$pV^k = \text{constant} \quad -\infty \leq k \leq \infty$$

$$p_i V_i^k = p_f V_f^k = \text{constant}$$

$$\int p dV = \text{constant} \int_i^f \frac{dV}{V^k} = \text{constant} \left(\frac{V^{-k+1}}{-k+1} \right)$$

$$\int p dV = \frac{\text{constant}}{1-k} (V_f^{1-k} - V_i^{1-k}) = \frac{p_f V_f^k V_f^{1-k} - p_i V_i^k V_i^{1-k}}{1-k}$$

$$\int p dV = \frac{p_f V_f - p_i V_i}{1-k} \quad \text{Valid for any } k \text{ except } k = 1$$

For ideal gas:

$$\int p dV = \frac{n\bar{R}T_f - n\bar{R}T_i}{1-k} = \left(\frac{n\bar{R}}{1-k} \right) (T_f - T_i)$$

Frictionless Quasistatic Adiabatic Process (Reversible Adiabatic)

Ideal gas with constant specific heats

$$k = \gamma = \frac{c_p}{c_v} \quad \int p dV = \frac{p_f V_f - p_i V_i}{1 - \gamma} \quad \gamma = \text{constant}$$
$$\int p dV = \frac{n\bar{R}T_f - n\bar{R}T_i}{1 - \gamma} = \left(\frac{n\bar{R}}{1 - \gamma} \right) (T_f - T_i)$$

$$W_{i-f} = -(n\bar{C}_v)(T_f - T_i)$$

Ideal gas with constant specific heats undergoing adiabatic process (reversible/irreversible)

$$Q_{i-f} = U_f - U_i + W_{i-f}$$
$$0 = (n\bar{C}_v)(T_f - T_i) + W_{i-f}$$

$$W_{i-f} = -(n\bar{C}_v)(T_f - T_i)$$

For the same end states

Hyperbolic Process

$$k = 1: \quad pV = \text{constant} = p_i V_i = p_f V_f$$

$$\int_i^f p dV = p_i V_i \int_i^f \frac{dV}{V} = p_i V_i \left(\ln \frac{V_f}{V_i} \right)$$

Ideal Gas: For an ideal gas hyperbolic process results in isothermal process

$$p_i V_i = p_f V_f = n \bar{R} T$$

For constant pressure (isobaric) process $k = 0$:

$$\int_i^f p dV = p \int_i^f dV = p(V_f - V_i)$$

Ideal gas: $W_{i-f} = n\bar{R} (T_f - T_i)$

For constant volume (isochoric) process $k = \infty$: $\int_1^2 p dV = 0$
 $p^{\frac{1}{\infty}} v = c$; hence $v = c$

Quasi-static isothermal expansion and compression of an ideal gas. The work is given by

$$p_i V_i = p_f V_f = n\bar{R}T \quad k = 0:$$

$$\int_i^f p dV = p_i V_i \int_i^f \frac{dV}{V} = p_i V_i \left(\ln \frac{V_f}{V_i} \right)$$

$$W = n\bar{R}T \int_i^f \frac{dV}{V} = n\bar{R}T \left(\ln \frac{V_f}{V_i} \right)$$

Quasi-static isothermal increase of pressure on a solid

$$dV = \left(\frac{\partial V}{\partial p} \right)_T dp + \left(\frac{\partial V}{\partial T} \right)_p dT$$

Since the **isothermal compressibility** is given by:

$$\kappa = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T \quad dV = -\kappa V dp$$

$$W = \int_{p_i}^{p_f} p dV = - \int_{p_i}^{p_f} p \kappa V dp$$

For constant temperature, κ and V are practically constant, so that may be taken out of the integrand

$$W = \int_{p_i}^{p_f} p dV = - \int_{p_i}^{p_f} p \kappa V dp = \frac{-\kappa V}{2} (p_f^2 - p_i^2)$$

Linear spring-loaded piston-cylinder

For the piston to be in mechanical equilibrium there should be no net acceleration $\Rightarrow m_p a = 0$

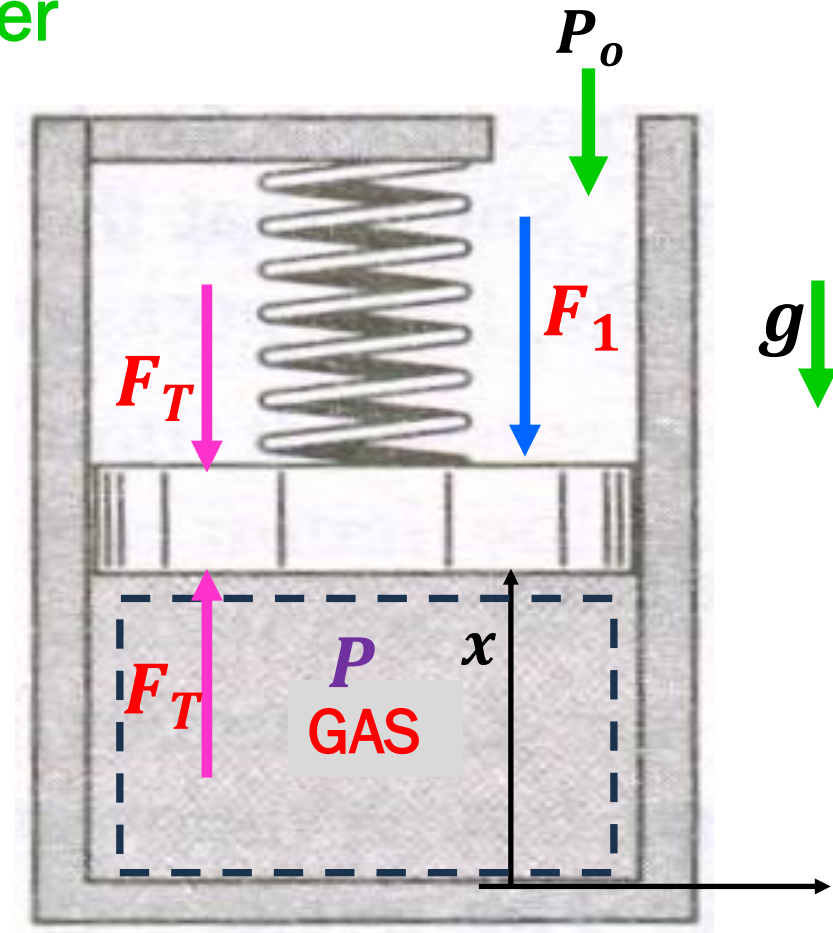
$$F_T \downarrow = F_T \uparrow$$

$$F_T \uparrow = PA$$

$$F_T \downarrow = P_o A + m_p g + k_s(x - x_o) + F_1$$

$$PA = P_o A + m_p g + k_s(x - x_o) + F_1$$

$$P = P_o + (m_p g + k_s(x - x_o) + F_1)/A$$



$$P = P_o + \left(\frac{m_p g + F_1}{A} \right) + \frac{K_s}{A^2} (V - V_o)$$

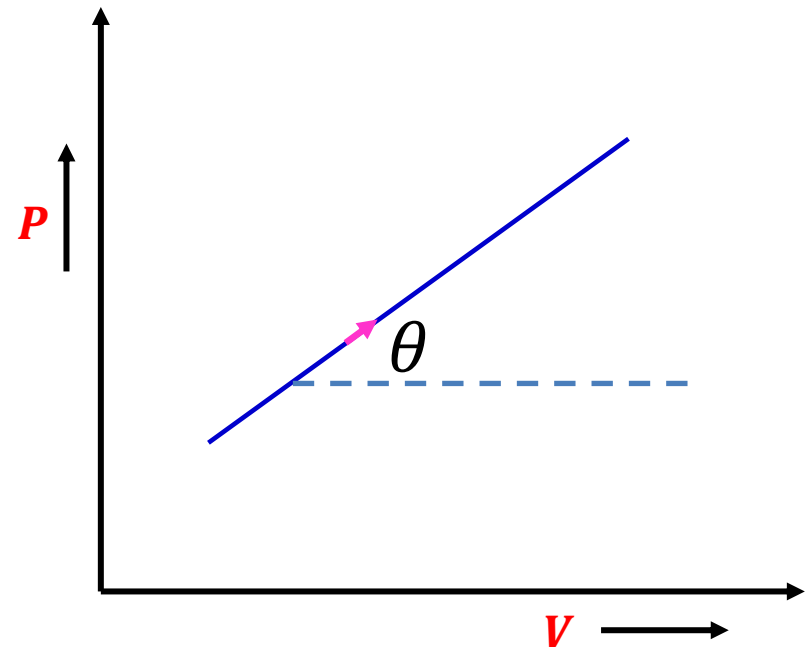
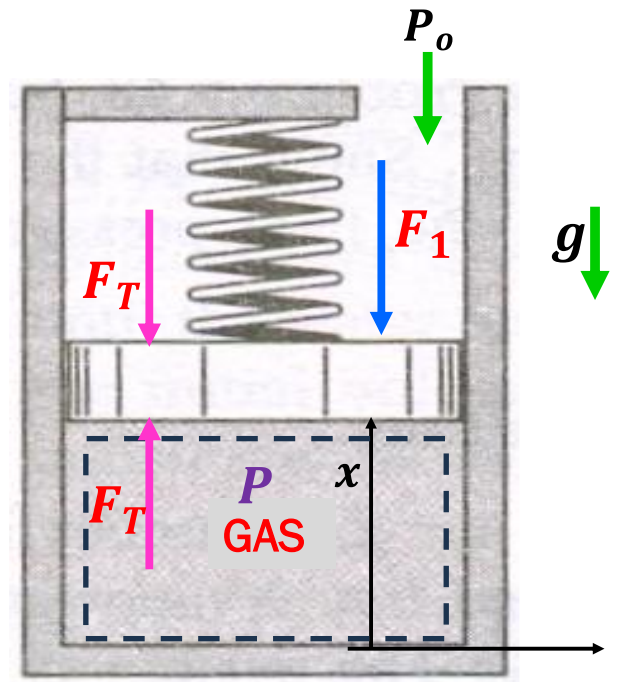
$$Ax = V$$

$$P = C_1 + C_2 V$$

$$C_1 = P_o + \left(\frac{m_p g + F_1}{A} \right) + \frac{K_s V_o}{A^2}$$

$$C_2 = \frac{K_s}{A^2}$$

P varies linearly with V
and the slope of the
 $P - V$ line is $\frac{K_s}{A^2}$



If such a system under goes a quasi-equilibrium process

$$W_{1-2} = \int_1^2 P dV = \int_1^2 (C_1 + C_2 V) dV = C_1(V_2 - V_1) + \frac{C_2}{2}(V_2 + V_1)(V_2 - V_1)$$

$$W_{1-2} = C_1(V_2 - V_1) + \frac{C_2}{2}(V_2 + V_1)(V_2 - V_1) =$$
$$\left(\frac{1}{2}\right)(V_2 - V_1)[(C_1 + C_2 V_2) + (C_1 + C_2 V_1)]$$

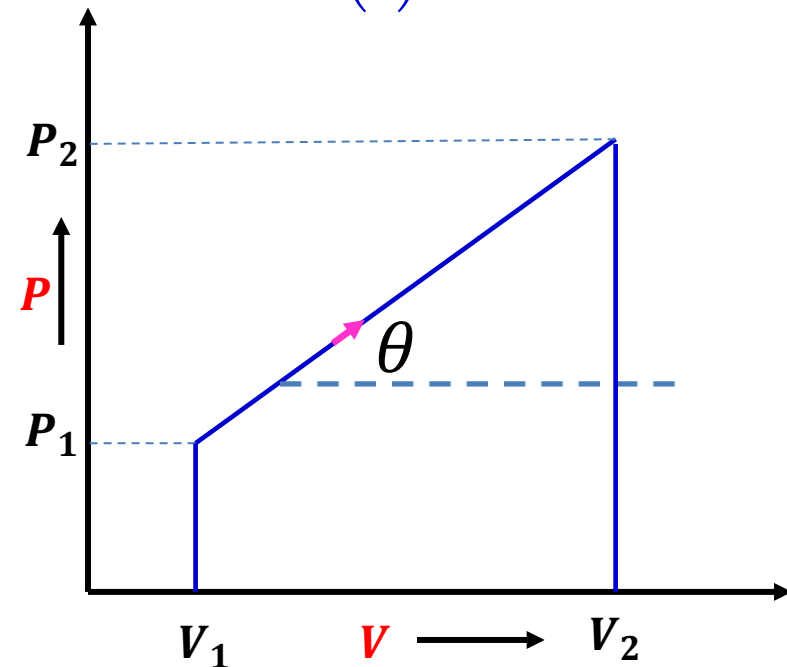
$$W_{1-2} = (V_2 - V_1) \left(\frac{1}{2}\right) [(C_1 + C_2 V_2) + (C_1 + C_2 V_1)] = (V_2 - V_1) \left(\frac{1}{2}\right) [P_1 + P_2]$$

Work done from area concept

$$W_{1-2} = \text{area of the trapezium}$$
$$= (V_2 - V_1) \left(\frac{1}{2}\right) [P_1 + P_2]$$

Work done from P_{ext}

$$W_{1-2} = \int_1^2 P_{ext} dV$$



Another Linear System

$$PA = P_o A + m_p g + m_w g$$

$$P = P_o + \left(\frac{m_p g + m_w g}{A} \right)$$

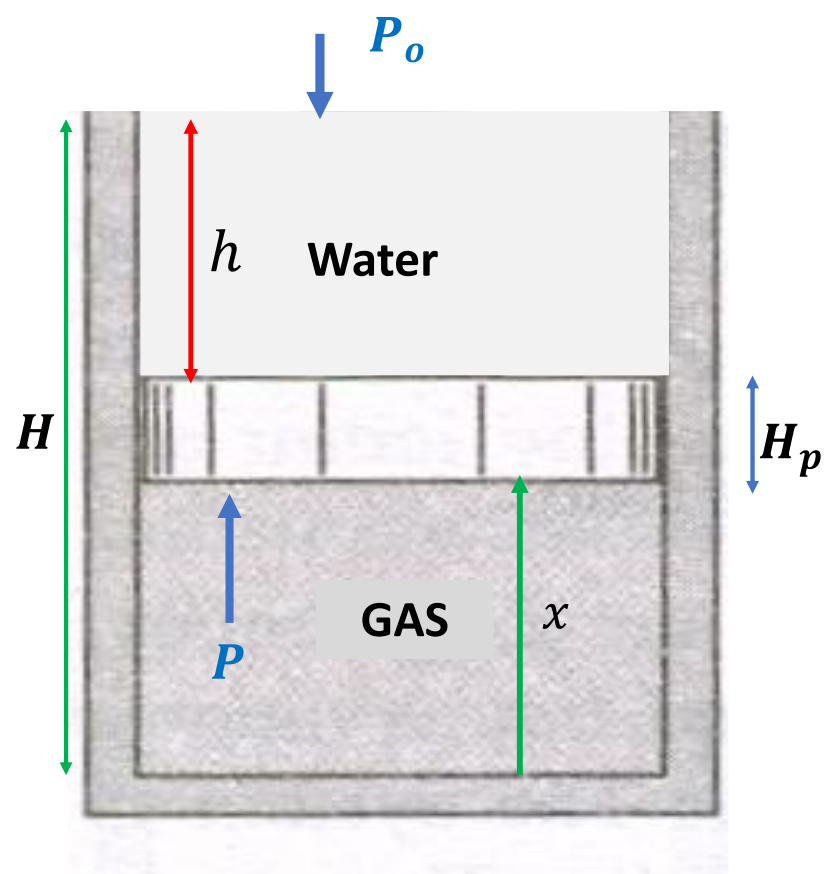
$$P = P_o + \left(\frac{m_p g + Ah\rho_w g}{A} \right)$$

$$P = P_o + \frac{m_p g}{A} + h\rho_w g$$

$$W_{1-2} = \int_1^2 P dV = \int_1^2 \left(P_o + \frac{m_p g}{A} + (H - H_p - x)\rho_w g \right) dV$$

$$W_{1-2} = \int_1^2 P dV = \int_1^2 \left(P_o + \frac{m_p g}{A} + \frac{A(H - H_p)\rho_w g - Ax\rho_w g}{A} \right) dV$$

$$W_{1-2} = \int_1^2 P dV = \int_1^2 (C_1 - C_2 V) dV \quad Ax = V \quad Adx = dV$$



Irreversible process with friction: Friction opposes relative motion

For expansion process: $\mathbf{PA = P_{ext}A + F \Rightarrow P = P_{ext} + F/A}$

For compression process: $\mathbf{PA = P_{ext}A - F \Rightarrow P = P_{ext} - F/A}$

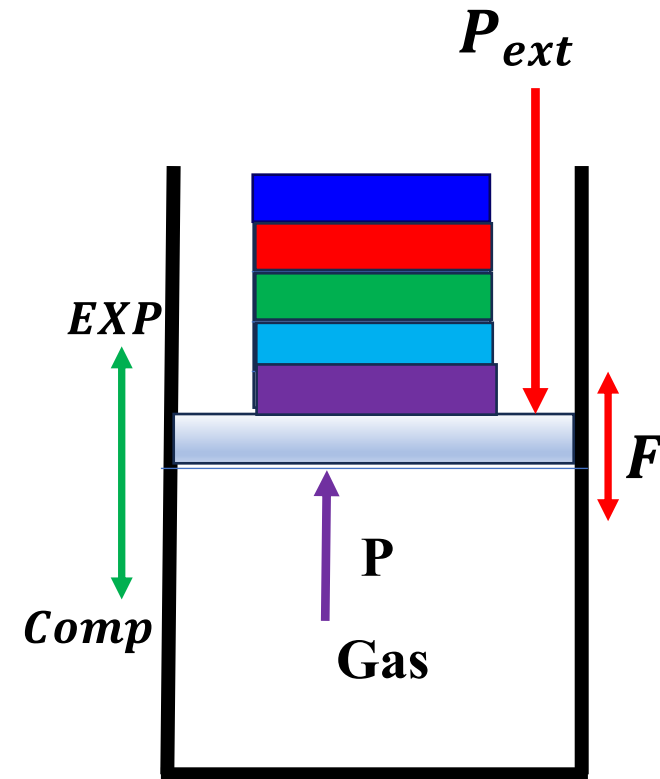
$$\mathbf{P = P_{ext} \pm F/A}$$

Actual Work done: $\delta W_{in} = P_{ext}dV$

During Expansion: $\delta W_{irr} = (P - F/A)dV$

During Compression: $\delta W_{irr} = (P + F/A)dV$

$$\mathbf{Lost\ Work: \int_1^2 P dV - \int_1^2 P_{ext} dV}$$



For the same displacement, **work done by the system during expansion will be maximum** and **work done on the system will be minimum during compression** if the process is **reversible**

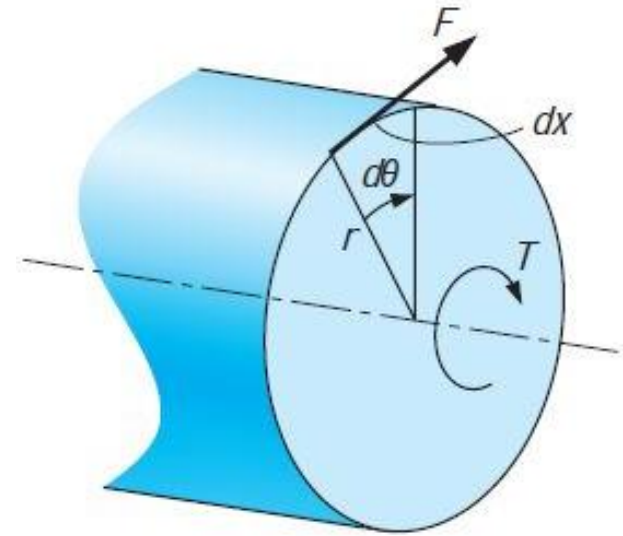
Shaft Work

Whenever a rotating shaft carrying a torque load crosses a system boundary, then rotating shaft work is done

$$(\delta W)_{\text{rotating shaft}} = -T d\theta$$

$$(W)_{1-2} = \int_1^2 -T d\theta$$

$$T = r \times F$$



Where T is the torque vector of the shaft and $d\theta$ is the angular displacement vector. *Normally thermodynamic problem statements do not require rotating shaft work to be calculated from above equation. The rotating shaft work is usually openly given as the part of the problem statement.*

$$\dot{W} = \int_1^2 -T \cdot \frac{d\theta}{dt}$$

$$\text{For Constant } T \quad \dot{W} = T\omega$$

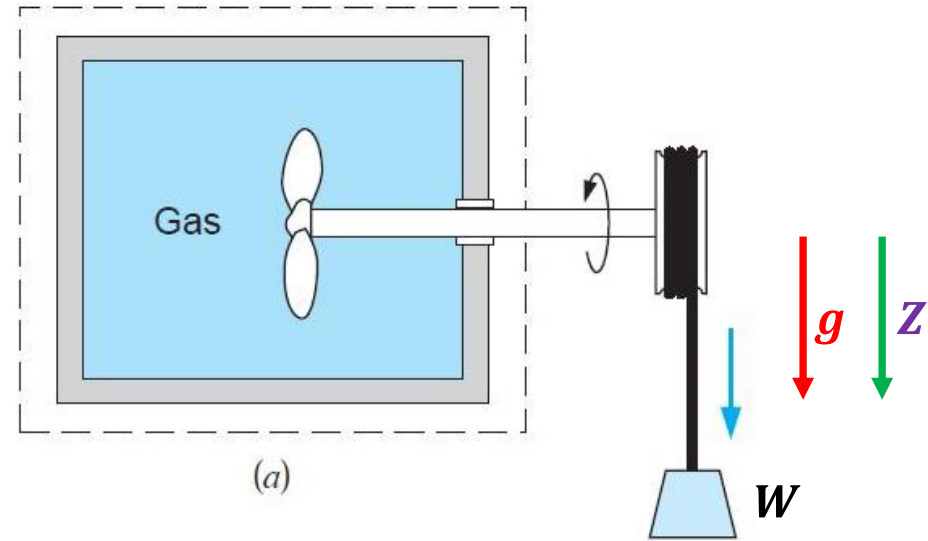
Paddle Wheel Work

There are work transfer to the system because weight is lowered. However

$$\int P dV = 0 \text{ as } V = \text{constant}$$

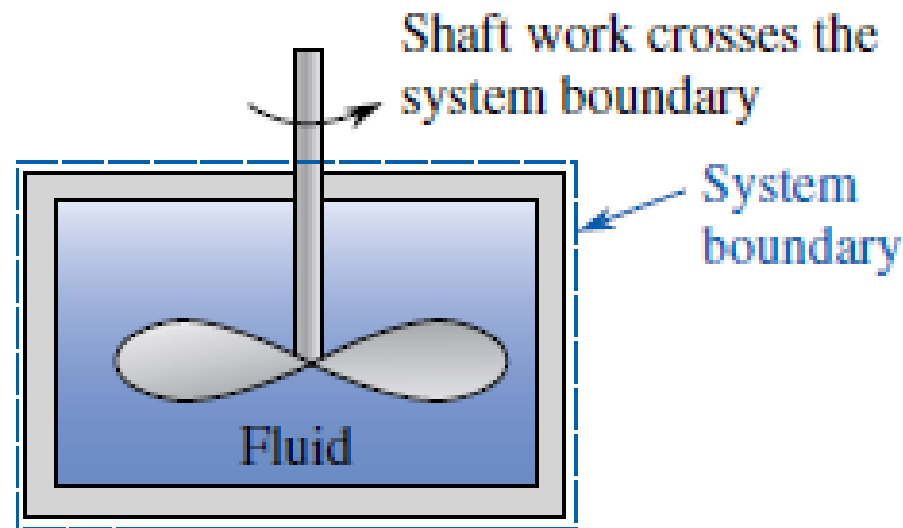
$$\delta W = (Mg) \cdot dZ = T \cdot d\theta$$

$$(W)_{1-2} = \int_1^2 (Mg) \cdot dZ \int_1^2 T d\theta$$

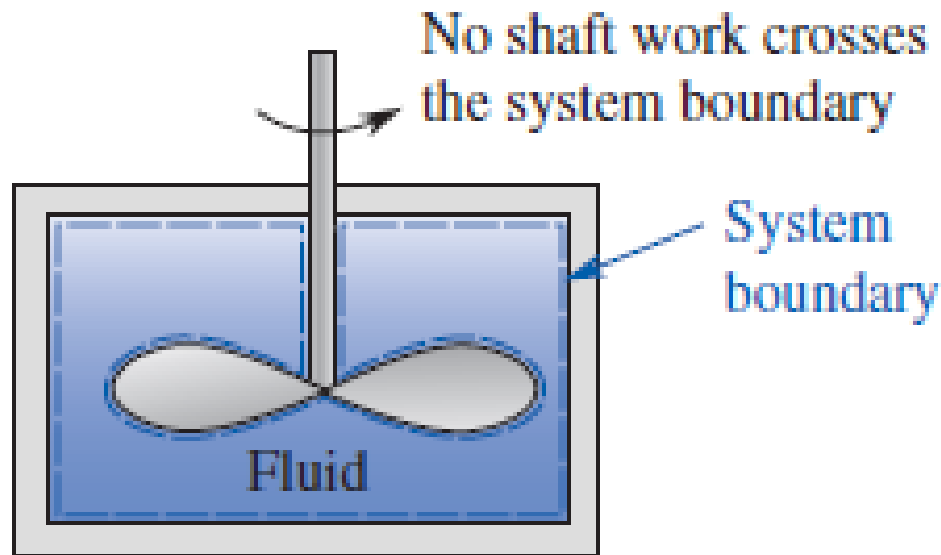


Some Insight on Shaft Work

Suppose you have a system that contains a fluid, and this fluid is in contact with a mixing blade or an impeller driven by a shaft passing through the system boundary (see Figure). **This would constitute an example of shaft work.** The shaft and the blade or impeller are inside the system and their physical and thermodynamic properties are part of the system's properties. **You have a heterogenous system made up of the fluid and the solid shaft and blade.** **If the mass of the fluid is large enough and the size of the shaft and blade is small enough, then their impact on the system's properties can be neglected and the system can be considered to consist of the fluid alone.** However, this is not always the case.



Suppose now you exclude the shaft and the blade or impeller from the system by restricting the system to be only the fluid and redraw the system boundaries so that they pass along the surface of the shaft and blade (see Figure). Now, your system consists of a pure substance (the fluid), but what kind of work mode do you now have? Since the only work modes we can analyze are “reversible,” the fluid medium cannot possess viscosity (fluid friction), and consequently, there can be no shear forces on the blade. The only force a viscous less fluid can exert on the blades is a pressure force, p . As the blade moves, the system boundary must move accordingly to keep up with it, and the pressure force on the blade must also move. This is just the definition of the moving boundary work mode. Consequently, this type of shaft work is not really shaft work at all, it is really moving boundary work.



Elastic Work

Whenever we **compress or expand an elastic solid** (like a spring) we perform elastic work. Consider a force $\pm F$ applied to the end of an elastic rod. The normal stress σ in the rod is

$$\sigma = \frac{\pm F}{A}$$

Where A is the cross-sectional area of the rod. Since the force $\mathbf{F}$ and its corresponding displacement $d\mathbf{x}$ are always in the same direction. Hence the vector dot product $\mathbf{F} \cdot d\mathbf{x}$ always reduces to Fdx where $F = |\mathbf{F}|$ and $d\mathbf{x} = |d\mathbf{x}|$

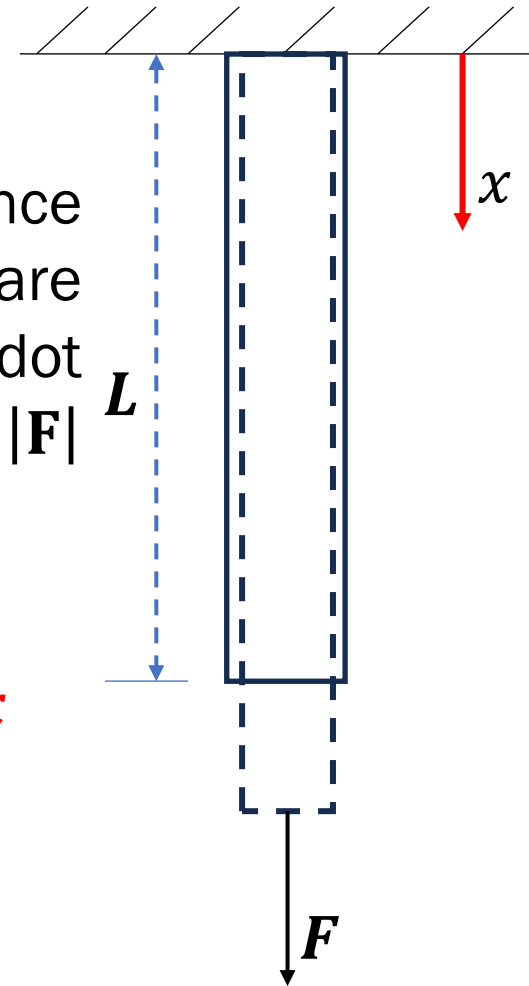
The work increment can be written as

$$\delta W = -\mathbf{F} \cdot d\mathbf{x} = -Fdx = -\sigma A dx$$

Now the strain ϵ in the rod is defined as

$$d\epsilon = \frac{dx}{L} = \frac{A dx}{AL} = \frac{dV}{V}$$

Where L is the length of the rod and AL is the volume V of the rod.



$$Adx = dV = Vd\epsilon$$

$$\delta W = -\mathbf{F} \cdot d\mathbf{x} = -Fdx = -\sigma Adx = -\sigma dV = -\sigma Vd\epsilon$$

$$W_{1-2,elastic} = - \int_1^2 \sigma V d\epsilon$$

Determine an expression for the work involved in deforming a constant volume elastic solid that obeys Hook's law of elasticity

$$\sigma = E\epsilon$$

$$W_{1-2,elastic} = - \int_1^2 E\epsilon V d\epsilon = -EV \int_1^2 \epsilon d\epsilon = -EV \left(\frac{\epsilon_2^2 - \epsilon_1^2}{2} \right) = -\frac{V}{2E} (\sigma_2^2 - \sigma_1^2)$$

The **-ve sign** is to take care of sign convention of work

Surface Tension Work

Surface tension work is the two-dimensional analog of the elastic work considered. Figure shows a soap film on a wire loop. One side of the loop has a movable wire slider that can either compress or extend the film. As in the case of an elastic rod the force and the deflection are always in the same direction and keeping the sign convention of thermodynamic work a – sign is incorporated.

Work in changing the area of a surface film

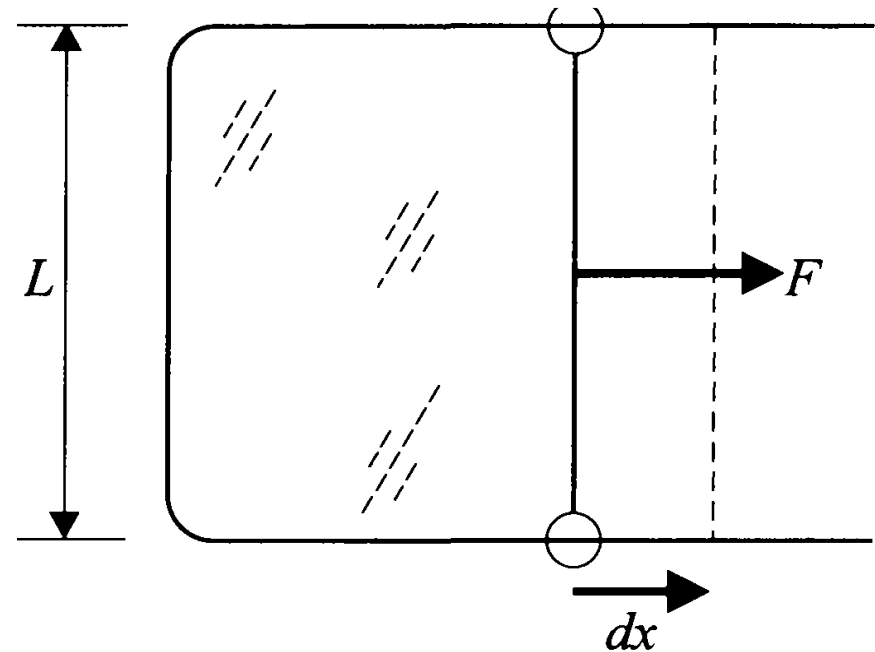
$$F = 2\gamma L$$

For an infinitesimal displacement dx

$$\delta W = -2\gamma L dx = -\gamma dA$$

$$W = - \int_{A_i}^{A_f} \gamma dA$$

For constant surface tension $W = -\gamma(A_f - A_i)$



A **quasi-static process** may be approximated by maintain the external force at all time only slightly different from the force exerted by the film

Soap bubbles is to be enlarged by blowing air into it, we want to **calculate work input required to enlarge the bubble**, which can be determined by the **change in area times the surface tension of the soap solution**



Determine the amount of surface tension work required to inflate a soap bubble from a diameter of zero to 0.05 m. The surface tension of the soap film can be taken to be a constant 0.04 N/m.

$$W_{1-2 \text{ (surface tension)}} = -\gamma \int_1^2 dA = -\gamma(A_2 - A_1)$$

$$A_1 = 0$$

$$A_2 = 2 \times (4 \times \pi \times R^2) = 0.0157 \text{ m}^2$$

$$W_{1-2 \text{ (surface tension)}} = -0.04 \left(\frac{\text{N}}{\text{m}} \right) \times (0.0157 - 0(\text{m}^2)) = -6.28 \times 10^{-4} \text{ J}$$